Fiscal Year:	FY 2021	**Task Last Updated:**	FY 12/13/2021
PI Name:	Snell, Edward H. Ph.D.		
Project Title:	Growth Rate Dispersion as a Predictive Indicator for Biological Crystal Samples Where Quality Can be Improved with Microgravity Growth		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Buffalo	**State:**	NY
Zip Code:	14203-1102	**Congressional District:**	26
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	01/24/2014	**End Date:**	01/23/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Grant extends to 1/23/2021 with new grant number 80NSSC19K1665; information per B. Penn/MSFC (Ed., 11/12/19) NOTE: End date changed to 1/23/2020 per NSSC information (Ed., 3/12/19)		
Key Personnel Changes/Previous PI:	November 2018 report: Tim Stachowski is a student at Roswell Park Comprehensive Cancer Center. Now that the initial flight samples have returned and X-ray analysis has begun Tim has been involved with the X-ray analysis and developing samples for the next flight. His samples were on orbit as part of LMMBIO-6. He has completed his PhD. and has moved to a Post Doctoral Fellow position at St. Judes Children's Hospital.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1665 ; NNX14AD80G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR's NOTE (11/12/2019): Grant extends to 1/23/2021 with new grant number 80NSSC19K1665; information per B. Penn/MSFC. Biological macromolecules make up the machinery, the instruction set, and the scaffold of life. They are smaller than the wavelength of visible light; thus, sophisticated techniques are needed to visualize them, and through this process, understand how life works. Structural understanding helps to discern mechanisms; once we understand how the machinery works, we aid, or more commonly impede that machinery through appropriate pharmaceutical design. The principal means to visualize the structure of these macromolecules is X-ray crystallography, a process that measures the diffraction of X-rays from ordered crystals to calculate the atomic structure. The importance of this process is exemplified in over 12 Nobel Prizes awarded to discoveries made from biological structures derived by X-ray crystallography. A limiting element to the process is the availability of high-quality, well-diffracting crystals. Quality is defined as a crystal that provides X-ray diffraction data of sufficient completeness and detail (resolution) to see the structure and hence understand the biology of the system. Biological macromolecules often take many days to grow crystals. Growth in reduced acceleration (commonly termed microgravity) on an orbiting spacecraft extends the physical quality of macromolecular crystals through a reduction in the mosaic spread (caused by slight misalignments of the molecules in the crystal lattice and used as a measure of long-range order) and an increase in crystal volume. Resolution (a measure of short-range order) is not directly affected by microgravity, but can still benefit with the correct experimental design to exploit the demonstrated improvements in long-range order. Microgravity growth can yield an improved quality crystal; however, not all samples improve from microgravity growth. If we could predict which samples could be improved by crystal growth in microgravity, then the true potential of this medium could be exploited in an efficient manner. It is our hypothesis that growth rate dispersion could be an indicator of crystals whose quality could be improved in microgravity. Growth rate dispersion is a phenomenon encountered in crystallization where seemingly identical crystals, produced from the same conditions, grow at different rates. We contend that large growth rate dispersion on the ground is indicative of a sample that should be improved by microgravity growth. In the case of inorganic crystals, a decrease in growth rate dispersion is correlated with an increase in physical crystal quality. The dispersion itself is thought to be caused by surface artifacts on the growing crystals that induce lattice imperfections. Gravity enables convective flows to develop in a crystallization solution. These flows are driven by density differences as molecules are depleted from the surrounding solution during nucleation and growth of the crystal. We plan to monitor growth rate dispersion of crystals grown on the ground and in microgravity, to determine if there is a correlation between the physical qualities of the resulting crystals with those measurements. We propose to use molecular biology techniques to alter the crystallization contacts and shift the growth rate dispersion properties of a single protein from low to high to test our predictive hypothesis. Finally we propose to extend the study to a selection of good and poorly diffracting crystals on the ground and confirm that those displaying high-growth rate dispersion on the ground are those that are improved on orbit and generate improved structural data when their quality is exploited.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	If our hypothesis proves to be correct, we will develop a generally applicable means to identify macromolecular crystals where quality will improve with microgravity growth. Biological macromolecules make up the machinery, the instruction set, and the scaffold of life. They are smaller than the wavelength of visible light; thus, sophisticated techniques are needed to visualize them, and through this process, understand how life works. Structural understanding helps to discern mechanisms; once we understand how the machinery works, we aid, or more commonly impede that machinery through appropriate pharmaceutical design. The principal means to visualize the structure of these macromolecules is X-ray crystallography, which accounts for 90% of the known structural information. It is a process that measures the diffraction of X-rays from ordered crystals to calculate the atomic structure. The importance of this process is exemplified by 12 Nobel Prizes awarded to discoveries made from X-ray crystallographic structures of biological macromolecules. Crystallography requires high-quality, well-diffracting crystals; crystallization is the most significant impediment to structure determination. Quality is defined as a crystal that provides X-ray diffraction data of sufficient completeness and detail (resolution) to see the structure and hence understand the biology of the system. In our high-throughput crystallization screening laboratory at the Hauptman-Woodward Medical Research Institute approximately 50% of proteins samples crystallize in at least one chemical condition are tested and of those, approximately 50% go on to be deposited within the protein data bank – a depository for atomic resolution structural data. To restate this, some 25% of the macromolecules that enter the laboratory provide crystals that are of insufficient quality for structural determination, and for the 25% of macromolecules that do provide a structure, higher-quality crystals would enhance the structural detail to enable a better understanding of the molecular function. Microgravity crystallization has the potential to improve crystal quality and thereby provide novel structures for the 25% of macromolecules that produced crystals of insufficient quality for structure determination. In addition, microgravity crystallization has the potential to improve the level of detail that can be observed in structures that have previously been resolved. It has become increasingly apparent that many of the failures to get structural data from crystals are not that they don't diffract, but that the diffraction is to very low resolution. A more ordered, physically higher quality crystal would increase the signal-to-noise of recorded data and extend this resolution to a point where structural information may be available, or potential pathways for improvement could be quantitatively evaluated.
Task Progress:	X-ray data has been collected from a number of T4 bacteriophage lysozyme studies and one structure has been deposited in the protein data bank, structural ID 4S0W. Three more structures will follow. The flight experiments suffered from significant delay and dehydration. This impacted harvesting and, eventually, crystals were harvested, cryogenically protected, and examined on the Industrial Macromolecular Crystallography Association (IMCA) line at the Advanced Photon Source. The data was not of high enough quality to make conclusive conclusions. The experience and results from growth rate studies are described in a comprehensive article "Microgravity as an environment for macromolecular crystallization – an outlook in the era of space stations and commercial space flight." Snell EH and Helliwell JR. (2021) Crystallography Reviews, 8th April, 1-44. (Ed. note: see Bibliography section.) Note that multiple funding sources covered the full period of research described in this paper and that an entire section of the paper described the research performed on the NASA grant. Research on some of the samples that did target microgravity growth, but were unsuccessful due to the dehydration, were published as part of related National Science Foundation-supported ground studies -- "Structural Insights Into

	Conformational Switching in Latency-Associated Peptide Between Transforming Growth Factor ß-1 Bound and Unbound States." Stachowski TR, Snell ME, Snell EH. 2020 Feb 6;7(Pt 2):238-252. https:// . eCollection 2020 Mar 1, and "Structural consequences of transforming growth factor beta-1 activation from near-therapeutic X-ray doses." Stachowski, T, Grant, TD, Snell, EH. Journal of Synchrotron Radiation 2019, 26, 967-979. https://
Bibliography Type:	Description: (Last Updated: 12/14/2021)
Articles in Peer-reviewed Journals	Snell EH, Helliwell JR. "Microgravity as an environment for macromolecular crystallization – an outlook in the era of space stations and commercial space flight." Crystallography Reviews. 2021;27(1):3-46. https://doi.org/10.1080/0889311X.2021.1900833 , Apr-2021